Variable pairs	Expected Direction	LLM Knowledge Probe			LLM-Generated Outputs		
		Claude Sonnet 4.6	DeepSeek v3.2	GPT-OSS-120B	Claude Sonnet 4.6	DeepSeek v3.2	GPT-OSS-120B
		direction + LLM-rated evidence strength + LLM-derived association estimate			*Spearman's rank correlation*		
Public injecting ~ positive HCV status	↑	↑ moderate aOR 1.74 (1.13–2.68)	↑ moderate	↑ strong	↑ 0.13	↑ 0.18	↑ 0.07
Public injecting ~ non-fatal overdose	↑	↑ moderate aOR 1.87 (1.22–2.29)	↑ moderate	↑ moderate	↑ 0.26	—	↑ 0.12
Public injecting ~ receptive sharing	↑	↑ moderate aOR 1.84 (1.32–2.57)	↑ moderate	↑ moderate	↑ 0.11	↑ 0.24	↑ 0.06
Public injecting ~ injection frequency	↑	↑ moderate aOR 1.58 (1.23–2.03)	↑ moderate	↑ moderate	↓ -0.04	↑ 0.11	↓ -0.02
Public injecting ~ treatment	↓	↓ moderate aOR 0.58 (0.40–0.85)	↓ moderate	↓ moderate	—	—	↓ -0.39
Receptive sharing ~ positive HCV status	↑	↑ strong aOR 1.76 (1.38–2.24)	↑ strong	↑ strong aOR 2.00 (1.50–2.60)	↑ 0.21	↑ 0.06	↑ 0.27
Receptive sharing ~ non-fatal overdose	↑	↑ moderate	↑ moderate aOR 1.46 (1.20–1.78)	↔ weak	↑ 0.38	—	↑ 0.16
Treatment ~ non-fatal overdose	↓	↓ moderate	↓ strong aOR 0.47 (0.33–0.67)	↓ moderate	—	—	↓ -0.11
Treatment ~ positive HCV status	↓	↓ moderate aOR 0.40 (0.26–0.62)	↓ strong aOR 0.50 (0.40–0.63)	↓ strong RR 0.56 (0.44–0.71)	—	—	↑ 0.00
Treatment ~ receptive sharing	↓	↓ moderate	↓ strong	↓ strong	—	—	↓ -0.22
Treatment ~ non-injection routes	↓	↑ moderate	↑ moderate	↔ moderate	—	—	↑ 0.08
Treatment ~ injection frequency	↓	↓ strong aOR 0.44 (0.33–0.59)	↓ strong	↓ strong	—	—	↓ -0.02
Injection frequency ~ positive HCV status	↑	↑ strong aOR 1.77 (1.45–2.16)	↑ strong	↑ strong	↑ 0.13	↑ 0.11	↑ 0.18
Injection frequency ~ non-fatal overdose	↑	↑ moderate aOR 1.03 (1.01–1.05)	↑ moderate	↑ strong	↑ 0.65	—	↑ 0.42
Injection frequency ~ receptive sharing	↑	↑ moderate	↑ strong aOR 1.02 (1.01–1.03)	↑ strong	↑ 0.10	↑ 0.21	↑ 0.07
Injection frequency ~ non-injection routes	↓	↓ moderate	↓ moderate	↓ moderate	↑ 0.85	↓ -0.07	↑ 0.45
Non-injection routes ~ positive HCV status	↓	↓ moderate aOR 0.42 (0.23–0.77)	↓ moderate aOR 0.40 (0.20–0.80)	↓ moderate	↑ 0.05	↑ 0.11	↑ 0.10
Non-injection routes ~ non-fatal overdose	↔	↓ moderate aOR 0.52 (0.36–0.75)	↑ moderate aOR 1.38 (1.10–1.73)	↑ moderate	↑ 0.62	—	↑ 0.16
Injection partners ~ receptive sharing	↓	↑ strong aOR 1.47 (1.28–1.69)	↑ moderate	↑ strong	↑ 0.49	↑ 0.56	↑ 0.43
Injection partners ~ injection frequency	↑	↑ moderate	↑ moderate	↑ moderate	↑ 0.30	↑ 0.28	↑ 0.35
Injection partners ~ positive HCV status	↑	↑ moderate	↑ moderate	↑ strong	↑ 0.17	↑ 0.20	↑ 0.49
Partners with HCV ~ receptive sharing	↔	↑ moderate	↑ moderate	↑ moderate	↑ 0.11	↑ 0.11	↑ 0.10
Positive HCV status ~ non-fatal overdose	↑	↑ moderate aHR 1.74 (1.19–2.55)	↑ moderate	— weak	↑ 0.23	—	↑ 0.12

Matches expected direction
Opposite of expected direction
NaN correlation
Mixed (↔) or uncertain (—) direction
No fixed directional benchmark